

The 2021 Puzzle

Andrew Juang

The 2021 Puzzle

Use all of the digits of 2021 (2,0,2,1) to write mathematical expressions that give results for the numbers 1-27.

Rules:

You can use the arithmetic operations (addition, subtraction, multiplication, division), and you can use factorial and exponentiation. You are allowed to concatenate digits (e.g., $41=20+21$), but not allowed to do something like $(2+1)0 = 30$. You may only use each of the digits once. Try to find as many solutions as you can.

Example: To create 1, you can do $2*0*2+1 = 1$.

1 =
2 =
3 =
4 =
5 =
6 =
7 =
8 =
9 =
10 =
11 =
12 =
13 =
14 =
15 =
16 =
17 =
18 =
19 =
20 =
21 =

$$22 =$$

$$23 =$$

$$24 =$$

$$25 =$$

$$26 =$$

$$27 =$$